

Policies and Permissions

1. Create a custom JSON-based policy in AWS IAM that allows read-only access to all objects in an S3 bucket named 'my-music-playlists', and attach this policy to a new user called 'playlistViewer'.

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1. Log in to the **AWS Management Console**.
 2. Open **IAM** → **Policies**.
 3. Click **Create Policy**.
 4. Select the **JSON** tab.
 5. Paste the code that you have to run :
 6. Save the policy with the name:
 7. Create a new IAM user named:
 8. Attach the policy to the user.

➤ **Result:** The user can list the bucket and read objects but cannot upload, delete, or modify files.
2. Write an inline policy in JSON that allows a user to start and stop EC2 instances only with the tag Environment:Dev, and attach it directly to an existing IAM user named 'devOpsIntern'.

***Hint:** Use 'Condition' in your policy to filter by tag.*

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- Attach the following **inline policy** to the IAM user **devOpsIntern**.
 - **Result:** The user can start and stop only EC2 instances tagged:
 - Environment = Dev
3. Create a managed policy in AWS IAM that grants permission to list and view DynamoDB tables, then attach this policy to a group called 'appReviewers' which could represent users reviewing app data like Zomato order history.

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- Create a managed policy using the following JSON:
 - Policy Name:
 - Attach this policy to the IAM group:
 - appReviewers
 - **Result:** Members of the **appReviewers** group can view DynamoDB tables and read data but cannot create, update, or delete records.

4. Given the following JSON policy, identify and fix the error that would prevent it from granting access as intended:
 { "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Action": "s3:GetObject", "Resource": "arn:aws:s3:::user-uploads/*" }] }
 Hint: Check if the policy allows listing the bucket as well as getting objects.

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❖ Issue

- The policy allows users to download objects but **does not allow them to list the contents of the bucket**. Without the s3:ListBucket permission, users may not be able to browse the bucket in the AWS Console.
5. Use ChatGPT to generate a JSON IAM policy that allows users to only update their own profile information in a DynamoDB table named 'UserProfiles', then review and test the policy in the AWS Policy Simulator.
 Hint: Ask ChatGPT for a policy with a condition that enforces users can update only their own data.

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❖ Explanation

- **GetItem** allows the user to view their own profile.
- **UpdateItem** allows the user to update their own profile.
- **Resource** limits access to the **UserProfiles** DynamoDB table.
- **dynamodb:LeadingKeys** ensures users can access only the item whose partition key matches their AWS username.

❖ Testing in AWS Policy Simulator

1. Open **IAM** → **Policy Simulator**.
 2. Select the IAM user.
 3. Attach or choose the policy.
 4. Test:
 - **GetItem** on the user's own profile → **Allowed**.
 - **UpdateItem** on the user's own profile → **Allowed**.
 - **UpdateItem** on another user's profile → **Denied**.
- This confirms that the policy enforces access only to the user's own profile.